

✓ Homework 3 Part 1

This is an individual assignment.

Solve all problems by hand. You may type your answers in markdown cells or  a PDF file with your handwritten answers.

✓ Problem 1 (10 points)

Consider a binary classification task where each sample $\mathbf{x}_i$ is d -dimensional with its corresponding label $t_i \in \{0, 1\}$. Suppose you have a dataset of N i.i.d. samples, $\{(x_i, t_i)\}_{i=1}^N$. Let $\mathbf{m}_i$ be the sample average for class C_i , $\mathbf{m}_i \in \mathbb{R}^d$.

The goal is to train a discriminant function of the form

$$y(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0$$

to separate the two classes. Recall that the vector $\mathbf{w}$ is orthogonal to the discriminant function $y(\mathbf{x})$ and that we are only interested in its direction not magnitude, thus the solution for $\mathbf{w}$ must satisfy the (equality) constraint $\mathbf{w}^T \mathbf{w} = 1$. Answer the following questions:

1. (8 points) Suppose that you are interested in finding $\mathbf{w}$ and w_0 such that the projected class means are the most separable. Write down the objective function, the Lagrangian function, and derive for the analytical solutions for $\mathbf{w}$ and w_0 .

1. we have 2 class, $m_1 = \frac{1}{N_1} \sum_{x \in C_1} x_i$, $m_2 = \frac{1}{N_2} \sum_{x \in C_2} x_i$

$$\max_w (w^T m_1 - w^T m_2)^2, \text{ s.t. } w^T w = 1$$

let $d = m_1 - m_2 \in \mathbb{R}^d$, $\rightarrow (w^T m_1 - w^T m_2)^2 = (w^T d)^2 = w^T (d d^T) w$.

$$\mathcal{L}(w, \lambda) = w^T (d d^T) w - \lambda (w^T w - 1)$$

$$\frac{\partial \mathcal{L}}{\partial w} = 2(d d^T) w - 2\lambda w = 0$$

$$\Rightarrow (d d^T) w = \lambda w$$

$\because w^T w = 1 \quad \therefore w = \frac{m_1 - m_2}{\|m_1 - m_2\|}$

$$y(x) = w^T x + w_0, \quad \mu_1 = w^T m_1, \quad \mu_2 = w^T m_2$$

$$\frac{\mu_1 + \mu_2}{2} = \frac{w^T m_1 + w^T m_2}{2}$$

$$\Rightarrow \frac{w^T m_1 + w^T m_2}{2} + w_0 = 0 \quad \Rightarrow w_0 = -\frac{1}{2} w^T (m_1 + m_2)$$

2. (2 points) **How would you change your approach if it is known that the dataset contains outliers?**

1. Use robust location estimates instead of means.

Replace m_1, m_2 with something less sensitive, e.g. per-class medians (coordinatewise), trimmed means, or M-estimates. Then do exactly the same derivation with it.

2. Downweight suspected outliers. Give each sample a weight $0 < a_i \leq 1$ (smaller for outliers), compute weighted class centers, then form $w \propto m_1 - m_2$

✓ Problem 2 (7 points)

Recall that the soft-margin Support Vector Machine objective function is

$$C \sum_{n=1}^N \xi_n + \frac{1}{2} \|w\|^2$$

subject to the constraints

$$\begin{aligned} t_n y(x_n) &\geq 1 - \xi_n, n = 1, \dots, N \\ \xi_n &\geq 0, n = 1, \dots, N \end{aligned}$$

Answer the following questions:

1. (2 points) From the training set, which points are used to make predictions during the test stage? Explain your reasoning.
2. (5 points) C is a parameter that is set by the user. Describe the relationship between values of C and (1) the resulting SVM decision surface, (2) performance and (3) number of support vectors.

Used points: Only the support vectors — the training samples for which $t_n y(x_n) = 1 - \xi_n$ (on the margin or misclassified).

Reason: Because only those points have nonzero multipliers ($\alpha_n > 0$) and thus contribute to the decision function at test time.

Non-support vectors ($\alpha_n = 0$) do not appear in the prediction equation and have no influence on future predictions.

Aspect	Small (C)	Large (C)
Decision boundary	Smoother, wider margin (allows more errors)	More complex, narrower margin (fits)
Performance	May underfit but generalizes better	May overfit with lower training error
Number of support vectors	More (many points inside margin)	Fewer (only boundary points)

✓ Problem 3 (18 points)

Complete the following table for the classifiers listed below (3 points/classifier).

Under "Type" list: generative, discriminative or probabilistic discriminative.

Under "Convergence" list whether the convergence is guaranteed or any other conditions for the converged solution.

Classifier	Type	Mapper function	Objective Function
Naïve Bayes Classifier	Generative	$y(x) = \arg \max_c P(C = c) \prod_{j=1}^d P(x_j C = c)$	Maximize joint likelihood $P(x, C)$ (or log-likelihood)
Fisher's LDA	Generative	$y(x) = w^T x + w_0$	Maximize between/within-class variance
Perceptron	Discriminative	$y(x) = \text{sign}(w^T x + w_0)$	Minimize classification error (iterative update)
Logistic Regression	Probabilistic Discriminative	$y(x) = \sigma(w^T x + w_0)$	Minimize negative log-likelihood (cross entropy)
Hard-margin SVM	Discriminative	$y(x) = \text{sign}(w^T x + w_0)$	Maximize margin $\frac{1}{2} \ w\ ^2$ s.t. $t_i(w^T x_i + w_0) \geq 1$
Soft-margin SVM	Discriminative	$y(x) = \text{sign}(w^T x + w_0)$	Minimize $\frac{1}{2} \ w\ ^2 + C \sum_i \xi_i$

✓ Problem 4 (5 points)

Draw a Perceptron to implement the **NAND gate** $\overline{A \cap B} = \overline{A} \cup \overline{B}$ using the threshold activation function $\phi(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$. Specify the numerical values for the weights and bias.

A	B	NAND(A,B)	X	$\phi(x)$
0	0	1	1.5	1
0	1	1	0.5	1
1	0	1	0.5	1
1	1	0	-0.5	0

$w_1 = -1$
 $w_2 = -1$
 $b = 1.5$

$$y = \phi(-1 \cdot A - 1 \cdot B + 1.5), \quad \phi(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

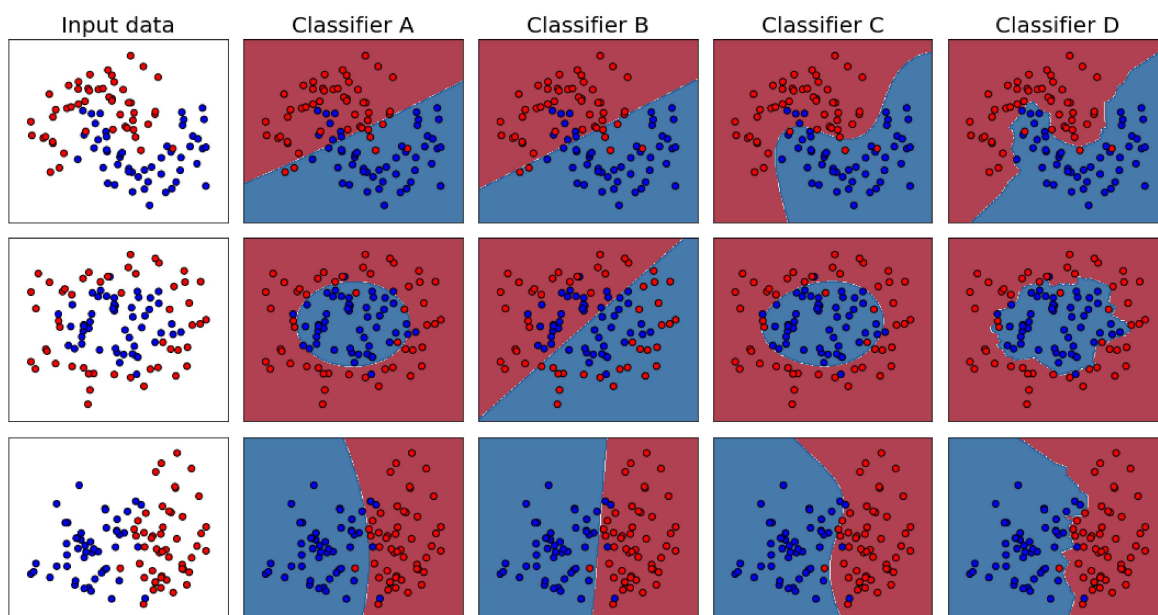
$A \rightarrow (-1)$
 $B \rightarrow (-1)$
 $\sum \rightarrow +1.5$
 $(bias)$
 $\phi(x) \rightarrow y$

✓ Problem 5 (5 points)

Consider the following performance from five different classifiers, A, B, C and D on three data sets. The classifiers used to produce these results are: Fisher's Linear Discriminant, k-NN with $k=3$, Naïve Bayes classifier, and SVM with RBF kernel.

Which classifier corresponds to which performance? Clearly justify your reasoning.

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1 from IPython.display import Image
2 Image('figures/Classifiers.png', width=900)
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Classifier B is LDA because in all three datasets it produces a single straight line decision boundary. LDA can only produce linear boundaries.

Classifier D is k-NN because its boundaries are very jagged/irregular and closely follow the data points in every dataset — a typical local, non-parametric behavior of k-NN with small k .

Classifier C is SVM with RBF because it makes smooth but flexible nonlinear boundaries: it perfectly bends around the two-moons data (first row) and makes a nice smooth closed contour for the donut data (second row). That's exactly what an RBF kernel SVM does.

Classifier A is then Naïve Bayes: its boundaries are parametric and simple — linear on the first set, axis-aligned elliptical on the donut set (because of the independent-Gaussian assumption), and a simple curved boundary on the third set. That matches Naïve Bayes with Gaussian features.

On-Time (5 points)

Submit your assignment before the deadline.

Submit Your Solution

Confirm that you've successfully completed the assignment